IT and Operations Miscoordination in a Supply Chain: Case of a Quick-Service Restaurant Chain

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While information integration is widely recognized as a key strategy for enhancing supply chain performance, our study uncovers a scenario where it results in unexpected repercussions. Using data from an independent purchasing cooperative that supplies a leading quick-service restaurant chain in the U.S., we delve into the effects of a Point of Sale (POS) system upgrade implemented across the chain. Counterintuitively, we observe a dip in customer satisfaction immediately following the new POS system upgrade. Concurrently, there is a noticeable uptick in order quantities placed by store managers. This dual phenomenon—rising orders and falling customer satisfaction—points to a potential misalignment between the POS upgrade and the ingredient procurement strategy. Our hypothesis finds support from three evident data patterns: (1) A decline in customer satisfaction concerning food quality following the POS upgrade. (2) An amplified decline in customer satisfaction when the POS system is adopted by a larger cluster of stores at once. (3) A short-lived duration of the negative effects resulting from the POS system upgrade. Furthermore, our analysis reveals that the surge in order quantities after the POS upgrade can be attributed to reduced uncertainties in demand forecasting affecting the newsvendor quantity. Our study highlights the nuanced implications of integrating advanced technologies into existing operational frameworks and calls for close collaboration between IT and operations business units.

1. Introduction

Over the past few decades, numerous companies have embarked on a journey of digital transformation, where technology on a large scale was leveraged to carve out a competitive edge. Within the realm of supply chain management, digital transformation has facilitated the process of the documentation and transfer of a vast amount of data. This not only takes place internally among

various departments but also among upstream and downstream partners within a supply chain. Integrating this information has been identified as a pivotal strategy to enhance supply chain performance (Kulp et al. 2004). A wealth of literature on information sharing within supply chains consistently underscores the considerable operational advantages of such information integration, as evidenced by Chen (2003) and others referenced therein.

Contrary to conventional wisdom, in this paper, we present a case where information integration leads to unintended outcomes. Our analysis draws from a dataset provided by an independent purchasing cooperative that functions as a supplier for a prominent quick-service restaurant chain in the U.S. This dataset encompasses intricate details of retail transactions at stores, replenishment orders between stores and distribution centers, and customer satisfaction metrics gathered from online surveys.

During our observation period, spanning from July 2013 to July 2015, our corporate sponsor progressively rolled out a cutting-edge Point of Sale (POS) system across the chain's outlets. In the legacy system, stores relayed their sales data in weekly aggregates to our partner firm. Using this data, recommended order quantities were then communicated back to store managers. As a result, these recommendations were calculated based on sales data that was approximately 10 days old, factoring in data transmission and processing time lags. However, the modernized POS system grants our partner firm instantaneous access to stores' sales data. With the new system, the *information lead time*—a concept introduced by Chen (1999)—was effectively reduced to zero. This shift in POS systems creates a unique natural experiment, allowing us to delve into the impact of information lead time on supply chain efficacy and customer satisfaction.

First and foremost, through both model-free evidence and a regression discontinuity analysis, we unveil an unexpected finding: customer satisfaction declines immediately following the POS system upgrade. In our primary analysis, customer satisfaction is quantified by the average number of surveys where the satisfaction score surpasses 7. The findings indicate that this metric decreases by approximately 0.152, which signifies a drop of roughly 7% when compared to the five-week average before the transition.

To discern the underlying reasons for customer dissatisfaction post-POS upgrade, we conduct a more detailed examination of the data. Specifically, our attention is drawn to fresh produce items most sensitive to operational shifts: cucumber, lettuce, spinach, and tomato. For all four ingredients, we observe a marked increase in order quantities by store managers after the POS switch. This trend, coupled with the decline in customer satisfaction, suggests a potential misalignment between the adoption of the new POS system and the strategy for ingredient procurement. It appears

the company failed to anticipate the amplified demand from stores after the POS change and consequently did not recalibrate its ingredient sourcing strategy to accommodate this shift. This hypothesis is corroborated by three distinct data trends: (1) There is a marked decline in customer satisfaction related to food quality following the POS upgrade. (2) The drop in customer satisfaction becomes even more pronounced when a greater number of stores transition to the new POS system simultaneously. (3) The negative impact of the POS upgrade is transient.

In our final analysis, we aim to understand the rationale behind store managers' increased orders following the POS upgrade. We present two competing hypotheses: (1) The first centers on the algorithmic perspective. Traditionally, store managers order ingredient supplies based on recommendations from the company headquarters, which are derived from a back-end algorithm. Essentially, this algorithm addresses a data-driven newsvendor problem. With the POS upgrade, demand forecasting uncertainty diminishes, subsequently influencing the newsvendor quantity. Historical data on ingredient supply and demand indicate a critical fractile below 0.5. Consequently, a decrease in demand uncertainty would logically result in heightened order quantities. (2) The second hypothesis revolves around human behavior. Store managers inherently aim to maintain consistent and timely ingredient supplies. However, anticipating potential disruptions during the POS system upgrade, they might proactively increase their orders as a buffer against any unforeseen issues. Determining the reasons for the increase in orders post-POS upgrade is vital for the company, as different root causes demand unique remedial actions. The observed data pattern lends support to the first hypothesis.

Our study offers a glimpse of the complexities of digital transformation and information integration within the context of supply chains. Information integration plays a crucial role in reducing information asymmetry among various stakeholders in a supply chain, a factor identified as fundamental in boosting supply chain effectiveness. Nonetheless, as firms undergo digital transformation, it is imperative to assess its broader operational impact and make necessary adjustments, as evidenced in our research. If not carefully managed, digital transformation could result in unintended consequences.

2. Literature Review

Our paper is related to the extensive literature on information sharing within supply chains. Different parties in a supply chain often make decisions independently, but the overall performance of a supply chain hinges on their coordinated decision-making. Endeavors to integrate information among different supply chain partners—through strategies like information sharing and synchronized replenishment—have been identified as key drivers to enhance supply chain efficacy (see, e.g.,

Chen 2003 and references therein).

While information sharing can be a two-way process, research primarily emphasizes the sharing of sales data from downstream players, such as retailers, to their supply chain partners. This focus stems from retailers' proximity to the market. Receiving timely sales data from these downstream members is widely regarded as an effective strategy to combat the bullwhip effect (Lee et al. 1997, Chen et al. 2000). Numerous studies have taken a modeling approach to examine the value of sharing demand information across a wide variety of supply chain structures (Gavirneni et al. 1999, Raghunathan 2001, Li 2002, Li and Zhang 2008, Ha et al. 2011, Cui et al. 2015, Shang et al. 2015, Wang et al. 2022, Jain 2022). The prevailing consensus in the literature highlights the substantial operational benefits of information sharing. Such sharing greatly aids in coordinating activities among various supply chain members, with the extent of improvement varying from modest single-digit enhancements to reductions nearing half of the total costs (Cachon and Fisher 2000, Lee et al. 2000, Sahin and Robinson Jr 2005). The difference in outcomes can be attributed to different assumptions regarding demand patterns, supply chain structures, and other factors. Beyond merely sharing information, firms should also ensure its prompt dissemination. Chen (1999) introduces the concept of "information lead time" and stresses the importance of different parties accessing accurate customer demand data promptly. Our research contributes to the existing literature by highlighting potential pitfalls: zero information lead time can be counterproductive if not aptly aligned with other supply chain functions.

Our study is also related to research centered on the integration of information technology (IT) within supply chains. Modern IT tools have streamlined the documentation and transfer of vast data sets across the supply chain. A wealth of studies underscores the benefits that organizations gain from the heightened visibility afforded by IT advancements. Such benefits include refined procurement procedures (Wu et al. 2007), strengthened partnerships within the supply chain (Sanders 2007, Paulraj et al. 2008, Aral et al. 2018), enhanced channel coordination (Gallino and Moreno 2014, Gallino et al. 2017), and elevated standards of service (Vickery et al. 2003), among others. The recent advancements in blockchain technology have garnered considerable attention due to their potential to significantly enhance transparency and improve traceability within the context of supply chain management (Cui et al. 2023a,b).

The two research streams converge when exploring the utilization or sharing of POS data to enhance operations among various supply chain entities. Parkan (2003) considers the implementation of a new POS system by a drugstore chain in Hong Kong across eight of its outlets. By analyzing cost and revenue data from these stores over an 18-month period, both pre and post-POS

system deployment, they demonstrate a marked improvement in operational efficiency attributed to the POS system's introduction. Steckel et al. (2004) explore the impact of shared POS information in a simulated "beer game." Their findings suggest that the benefits of sharing POS data are contingent on the demand pattern. POS information proves valuable when there's a shift in demand levels. However, its value diminishes when the demand remains stationary with cyclical patterns. On a related note, Williams et al. (2014) delve into strategies for upstream supply chain partners to leverage POS data from retailers. By integrating POS data with the retailer's order history, they show that forecasting accuracy can be enhanced, leading to improvement in the overall fulfillment planning processes.

Our paper contributes to the existing literature in two ways. First, quantifying the value of (instantaneous) POS information often proves challenging, as evidenced by Parkan (2003). Yet, this challenge diminishes in our unique context of the perishable food supply chain, as the quality of produce is intrinsically linked to supply chain operations. This enables us to discern the impact of POS information more clearly. Second, while most studies aim to quantify the value of POS information, in our context, this information is consistently accessible, albeit with a delay. Consequently, our primary emphasis in this paper centers on the influence of information lead time on a firm's daily operations and the ensuing impact on customer satisfaction. Intriguingly, our findings suggest that a reduction in information lead time does not unilaterally result in improved operations or heightened customer satisfaction. The realization of information integration's benefit requires coordination from the operations side of the organization.

Methodologically, our paper employs a regression discontinuity approach, a method that has received limited but steadily increasing attention in operations literature (Terwiesch et al. 2020). Noteworthy applications of regression discontinuity include studies on diverse topics: the impact of mandatory overtime laws on nursing home service quality (Lu and Lu 2017), the influence of regulatory oversight on delays and cost overruns in public infrastructure projects (Calvo et al. 2019), the effect of inventory-scarcity messages on online sales (Park et al. 2020), the consequences of tightening standards on automaker compliance (Hu et al. 2021), and the role of financial incentives in shaping driver responsiveness on a delivery platform (Zhang et al. 2023), among others. Our research is the closest related to a small body of literature that applies regression discontinuity design using time as the running variable, with a specific treatment date as the threshold. This technique allows for a consistent estimate of the average treatment effect, even when the assumption of random assignment into treatment and control groups is relaxed (Busse et al. 2006). In this context, Tan et al. (2017) focus on the movie rental industry, examining the impact of product

variety on demand distribution by analyzing data around the implementation of new agreements with suppliers. Similarly, Dhanorkar et al. (2021) explore the impact of expert service in online materials and waste exchanges by scrutinizing a narrow time window around the discontinuation of such services.

3. Data Description

The data was kindly supplied by the Wharton Customer Analytics Initiative (WCAI) from a corporate sponsor whose identity is withheld by request. The corporate sponsor is an independent purchasing cooperative that serves as a supplier to a major quick-service restaurant chain in the United States. The data observation window is from July 1, 2013, to July 1, 2015, including a total of 104 weeks and 3 days. Due to data irregularities at the beginning of and the end of the observation window, we remove the first week and last few days of observations in our analysis. As a result, the observation window of the selected sample is from July 8, 2013, to June 28, 2015.

The data set can be generally divided into three main categories: (1) Retail transactions collected from 4,116 stores, which spread across four geographic regions, namely west (California), southwest (Texas), northeast (Connecticut, New Jersey, New York, and Pennsylvania) and midwest (Illinois and Indiana). The transaction data details the specific menu items purchased, their respective quantities, and any discounts or promotions applied to the order. (2) Replenishment orders between stores and 245 (re)distributors. In our sample, each store primarily sources its ingredients from a single main distribution center. The replenishment data specifies the ingredients ordered, their respective quantities, and the suppliers for each ingredient. (3) Records of customer complaints collected through customer online surveys. In the surveys, customers provide feedback on their overall satisfaction with their orders, detailing any issues and specifying the affected products. Each customer survey record captures the date of the reported complaint and can be associated with specific stores, though not individual transactions. In our sample, there are 1,082,316 records from online customer surveys. Next, we discuss each of these three sources of data in detail.

3.1. Stores and Transactions

Our sample includes a total of 4,116 stores in the US. Geographically, they are distributed in four US regions, namely west (California), southwest (Texas), northeast (Connecticut, New Jersey, New York and Pennsylvania) and midwest (Illinois and Indiana). Stores also vary in their types. While the majority are standard stand-alone retailers, a small fraction are more compact and situated within other business establishments like convenience stores, gas stations, large supermarkets, restaurants, universities, and so on. Additionally, not all stores participate in the loyalty program, where customers have the opportunity to earn or redeem points on their purchases.

Table 1 Summary of store categories

	# of stores
Geographical regions (US)	
west	1,647
southwest	486
northeast	703
midwest	745
Store types	
traditional	3,202
others	379
Participation in loyalty program	
yes	923
no	2,658
Total # of stores	3,581

Given our research focus on examining the impact of information lead time on operations and customer satisfaction, we set forth the following criteria for selecting our sample. Our identification approach necessitates an extended observation period both before and after the POS implementation. Therefore, we exclude stores that have fewer than 5 weeks of observations on either side of the observation window. This leads to the removal of a total of 422 stores. Additionally, we exclude stores with missing information in either supply data or customer survey data within a 5-week time window before or after the POS implementation. As a result, we have a total of 3,581 stores in the sample for the analysis. General information of stores in the sample is summarized in Table 1.

We further illustrate some summary statistics of stores in Table 2. Average weekly sales per store is around $8,724. On average, a store is approximately 60 miles away from its primary distribution center. However, there are notable variations depending on the geographical locations of the stores. Most stores were well-established prior to the POS implementation, and thus, their sales patterns are anticipated to remain fairly stable over time.

3.2. Supply Data

Next, we provide a description of replenishment order data within our sample. To frame this discussion, it is essential to offer a brief overview of the business model of the quick-service restaurant chain we're examining. Each restaurant in this chain operates like a make-to-order workshop, where customers enjoy the flexibility to tailor their food orders based on their preferences, including choices of bread, meat produce, fresh vegetables, pickles, sauces, and so on. The inventory comprises 117 distinct items. These items span ingredients essential for food preparation, ready-to-consume

Table 2 Store statistics

	mean	std. dev.	min	max
Weekly sales ($)	8,724	3,633	1,271	34,526
Distance (mile)	60	47	0.5	263
Open time (day)	4,776	3,026	84	13,582

Note: This table summarizes the distribution of weekly sales, the distance between a store and its corresponding main distribution center, and the number of days between the store's first opening and the date when it implements the POS system.

products like cookies and bottled drinks, and ancillaries such as gloves and tissues. Regular replenishment of these items is crucial for smooth operations. Our research is anchored in understanding the impact of information lead time on operational decisions and customer satisfaction as a result. With this objective in mind, we zero in on ingredients that have a short shelf life due to their sensitivity to operational changes. Specifically, our analysis focuses on the following ingredients: cucumber, lettuce, spinach, and tomato.

Table 3 Supply of ingredients

	mean	std. dev.	min	max
Cucumber				
weekly supply (each)	5.66	3.53	0	75.43
# of transactions per week	1.08	0.53	0	4
Lettuce				
weekly supply (LB)	13.88	6.61	0.08	80.48
# of transactions per week	1.22	0.48	0	5
Spinach				
weekly supply (LB)	1.77	1.37	0.01	17.71
# of transactions per week	1.15	0.53	0	5
Tomato				
weekly supply (LB)	15.90	8.75	0.13	135.71
# of transactions per week	1.21	0.47	0	4

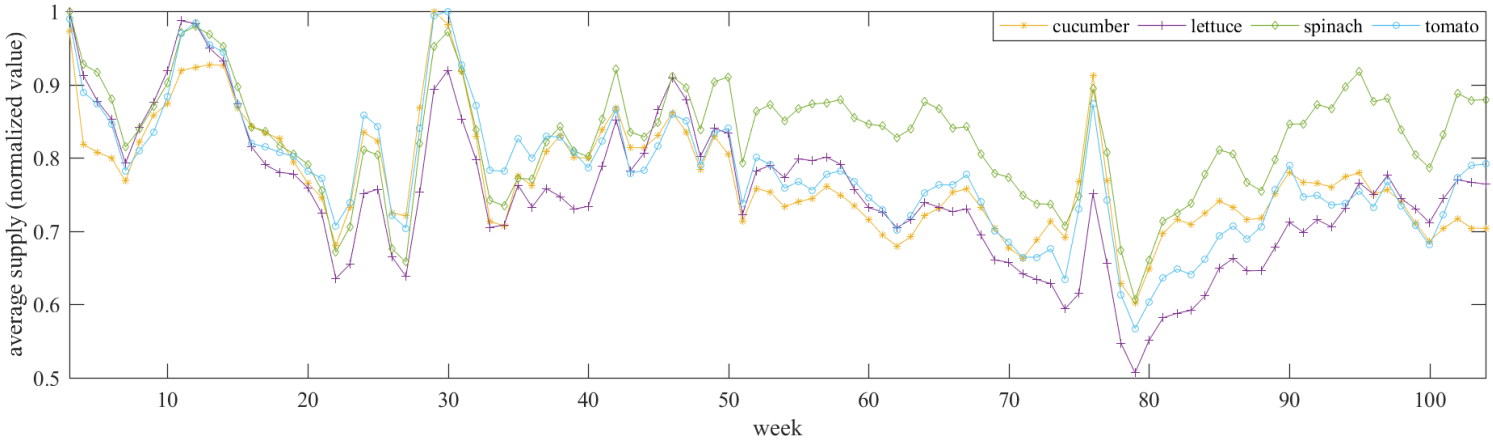
Note: We calculate the weekly supply based on the premise that ingredient consumption is uniformly distributed throughout the replenishment cycle. This allows us to address the challenge posed by the irregular intervals at which replenishment orders were placed.

The statistics detailing the number of transactions per week and the weekly supply for our chosen ingredients are presented in Table 3. Determining the number of transactions is straightforward from the sample. However, computing the weekly supply presents challenges due to the irregular intervals at which replenishment orders were placed. For our analysis, we assume that ingredient consumption is uniformly spread across these replenishment intervals. This allows us to distribute the supply quantities from each order over individual weeks. Table 3 suggests that, on average,

stores ordered a little more than once per week. As such, we aggregate our data at the weekly level in our analysis.

Figure 1 illustrates the evolution of the average weekly supply for each ingredient over time. While the supply levels for these ingredients generally remain stable, there is a noticeable seasonal pattern, with peaks typically occurring just before the Christmas holiday season.

Figure 1 Average weekly supply of ingredients over time



Note: To ensure anonymity, the weekly supply is normalized by the highest value found within the sample.

3.3. Customer Survey Data

Every customer is provided the opportunity to convey their satisfaction or dissatisfaction with the food and/or service they experienced. They can do this by completing an online survey, accessible via a link on their transaction receipt. The survey records customers' input regarding their overall satisfaction, as well as their satisfaction with food quality, speed of service, friendliness of employees, cleanliness of store, store's atmosphere, and so on. Each question is measured using a 10-point Likert scale, ranging from 0 = (very disappointed) to 10 (very satisfied). Each survey comes with a timestamp denoting the submission date and a restaurant ID, enabling us to associate survey responses with specific restaurants. However, it's important to note that we cannot directly link these surveys to individual transactions. To motivate customers to partake in the survey, the restaurant chain offers incentives such as potential discounts on future visits or modest rewards, like a free cookie, upon survey completion.

The two measures that are most relevant for our study are the overall customer satisfaction and customer satisfaction of food quality. Statistics of these two measures are summarized in Table 4.

There is a prevailing discussion about whether to categorize Likert scales as ordinal or quantitative, as referenced by Jamieson (2004) and Joshi et al. (2015). A prevalent viewpoint advocates for an ordinal interpretation, which we concur aligns better with our context as a fixed interval between points can be hard to justify. Consequently, we opt to use the frequency of scores instead of their mean as a metric to gauge customer satisfaction. Specifically, in our main analysis, we gauge customer satisfaction based on the number of surveys where the overall score exceeds a threshold of either 7 or 5. For robustness checks, we employ alternative metrics such as the average of these scores, and the results are presented in the Appendix.

Table 4 Statistics of customer satisfaction

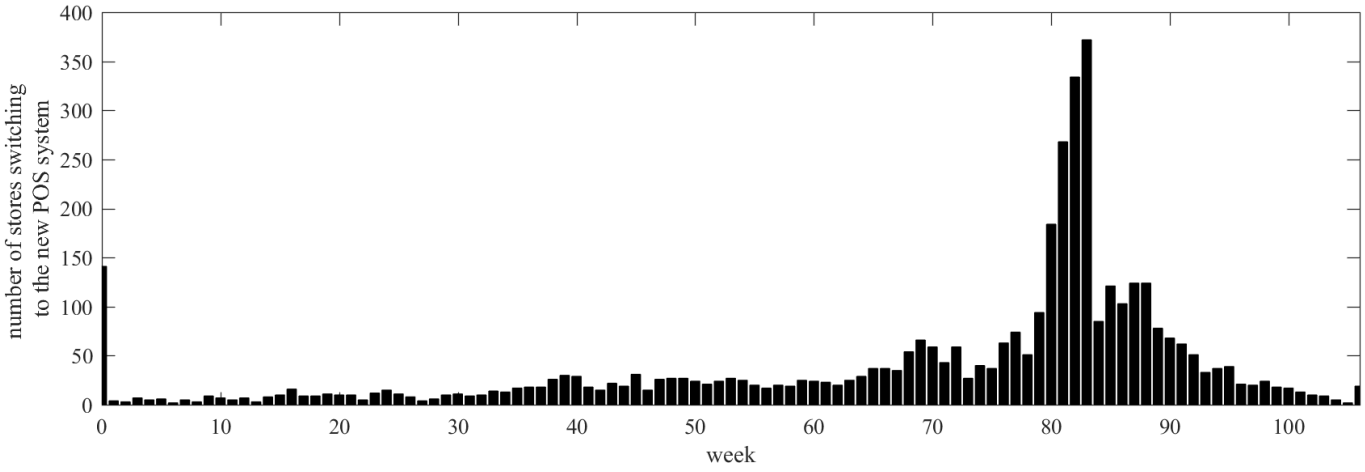
	Overall satisfaction		Food satisfaction	
Rating	Count	% in the sample	Count	% in the sample
0	18,096	1.67%	7,186	0.66%
1	6,845	0.63%	3,181	0.29%
2	6,964	0.64%	3,558	0.33%
3	7,213	0.67%	4,453	0.41%
4	7,885	0.73%	5,649	0.52%
5	24,524	2.27%	23,062	2.13%
6	19,851	1.83%	21,125	1.95%
7	45,444	4.20%	49,151	4.54%
8	102,437	9.46%	109,779	10.14%
9	165,922	15.33%	170,990	15.80%
10	677,135	62.56%	684,182	63.21%
Sample size	1,082,316			

3.4. Identification

During the observation window between July 2013 and July 2015, our corporate sponsor progressively introduced a new POS system, deploying it across the restaurant chain's outlets. Figure 2 illustrates the timeline of these installations. We refer to the systems pre and post-update as the "old" and "new" systems, respectively. Under the old system, stores transmitted their sales data in weekly batches to our corporate sponsor. Based on this data, recommended order quantities were then communicated to store managers. Consequently, these order recommendations were based on sales data from approximately 10 days prior, accounting for data transmission and processing delays. In contrast, the new POS system empowers our corporate sponsor with real-time access to sales data. That is, with the adoption of the new system, the information lead time was effectively reduced to zero.

The implementation of the new POS system presents a unique opportunity for us to examine the role of information lead time within supply chain operations. Generally, discerning the specific

Figure 2 Number of stores switching to the new POS system over time



Note: For completeness, the figure is created using the entire data sample of 4,116 stores.

impact of information or demand uncertainty in supply chain management is challenging due to numerous confounding variables such as general sales trends, competition, and so on. However, our dataset permits us to control for most of these typically unseen variables by narrowing our focus to a concise time frame surrounding the POS system's installation. Specifically, we analyze a window of 5 weeks before and 5 weeks after a store adopts the new system. One caveat is that sales data from individual stores only becomes available post-implementation of the new POS system. Consequently, our primary emphasis is on understanding the influence of reduced information lead time on both customer satisfaction and broader supply chain operations.

4. Impact of POS Switch on Customer Satisfaction

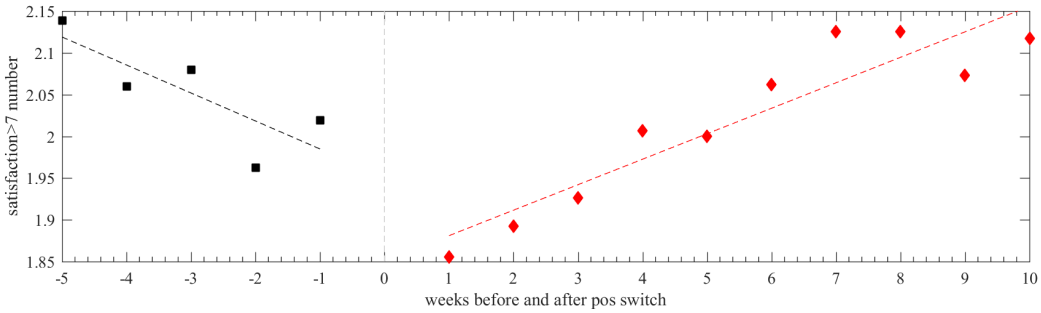
We first illustrate the impact of POS switch on customer satisfaction using both model-free evidence and results from a regression discontinuity study.

4.1. Model-Free Evidence

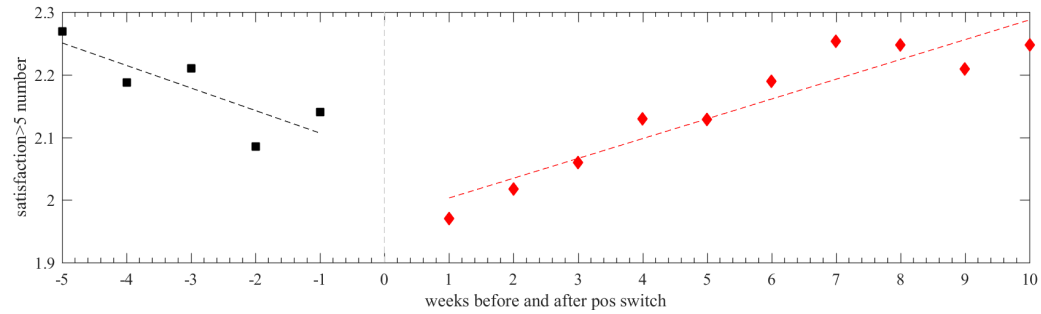
To gain insight into the potential effects of the POS transition, we begin by evaluating the raw data concerning customer satisfaction. We aim to identify any consistent patterns coinciding with the adoption of the new POS system in stores. As previously mentioned, a limitation of our dataset is the absence of transaction records prior to the POS switch. As a result, we cannot comment on any shifts in sales patterns attributable to the new system. One could argue that the switch to a new POS system should not noticeably impact sales, as it is a backend transition that remains largely invisible to customers.

To understand how customer satisfaction varies around the times of POS switch, we realign the

Figure 3 Customer satisfaction during the weeks around POS switch.



(a) Average number of surveys with overall satisfaction > 7



(b) Average number of surveys with overall satisfaction > 5

Note: The x-axis indicates the week. The y-axis in panel (a) (resp., panel (b)) denotes the average number of surveys with an overall satisfaction score > 7 (resp., > 5) per week. Dotted lines are fitted values from first-order polynomial regressions on either side of the time interval of the POS switch.

time periods in data by the week when the POS was switched for each store and plot the average number of surveys with an overall satisfaction score greater than 7 (resp., greater than 5) per week in Figure 3(a) (resp., Figure 3(b)). Interestingly, we observe a decline in overall customer satisfaction leading up to and immediately following the POS transition, hitting its lowest point during the first-week post-switch. However, this trend then reverses and begins to rise. The average number of surveys with a customer satisfaction score greater than 7 during the first week after the POS switch is 1.85, which is about 10% less than the average over the five weeks preceding the switch. This decrease is arguably very substantial.

4.2. Base Model

Next, we use a regression discontinuity approach to establish the impact of POS switches on customer satisfaction formally. To control for the unobserved store heterogeneity, we apply a store fixed-effects model with the following specifications:

$$Y_{it} = \alpha_i + \beta POS_{it} + \boldsymbol{\theta} \boldsymbol{X}_{it} + \epsilon_{it}, \tag{1}$$

Table 5 Regression results on overall customer satisfaction

	Overall satisfaction > 7	Overall satisfaction > 5
POS	-0.152^{***}	-0.156^{***}
	(0.045)	(0.047)
promotion	0.002	−0.029
	(0.164)	(0.165)
opentime	−0.000	−0.000
	(0.001)	(0.001)
Month fixed effects	Yes	Yes
Store fixed effects	Yes	Yes
Number of observations	35,810	35,810
Number of stores	3,581	3,581
R^2	0.638	0.643
Adj. R^2	0.597	0.603

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The regression analyses utilize data spanning 5 weeks prior to and 5 weeks following the POS switch.

where the dependent variables Y_{it} represents the number of customer surveys satisfying specific criteria for store i during week t. We present the main results with Y_{it} measuring the number of surveys with customer satisfaction that exceeds either 7 or 5. Regression results derived from alternative selection criteria, such as the average customer satisfaction scores, are presented in the Appendix.

In terms of the independent variables in the model, α_i captures the store fixed effect. POS_{it} is a binary independent variable, indicating whether store i has switched to the new POS system in week t. In particular, POS_{it} equals 1 if store i is using the new POS system in week t, and 0 otherwise. $\boldsymbol{X_{it}}$ includes a vector of control variables, which allows us to control for time-varying confounding factors. These variables include (1) A dummy variable $promotion_{it}$, which is equal to 1 if store i is running a promotion in week t, and 0 otherwise. To clarify, there are two types of promotions in the dataset: one "standardized" promotion, such as the "deal of the day," which recurs almost daily across all stores, and another type of limited-time offers that are tailored to particular menu items. The dummy variable $promotion_{it}$ is an indicator for the second type of limited-time offers, as the first type of promotion has the same effect both before and after the POS switch; (2) Monthly dummy variables $month_t$. Controlling for month-of-the-year fixed effects is important because there is clear seasonality in sales, as suggested in Figure 1.

The impact of the POS switch on customer satisfaction is summarized in Table 5. The outcomes from the base model align with our observations from Figure 3, indicating a notable decline in customer satisfaction following the POS switch. Specifically, there's an average reduction of 0.152 in the number of surveys per store per week recording overall satisfaction scores above 7 and a

0.156 decrease for scores above 5 when compared to the weeks leading up to the POS transition. For context, in the 5 weeks prior to the POS switch, the average number of surveys registering overall satisfaction scores above 7 and 5 were 2.06 and 2.19, respectively. This data suggests that the POS switch resulted in approximately a 7% drop in customer satisfaction. Conversely, factors such as limited-time promotions and the experience of franchise owners (measured by the number of days since the store's opening) do not appear to influence customer satisfaction.

5. Rationale for Customer Dissatisfaction After POS Switch

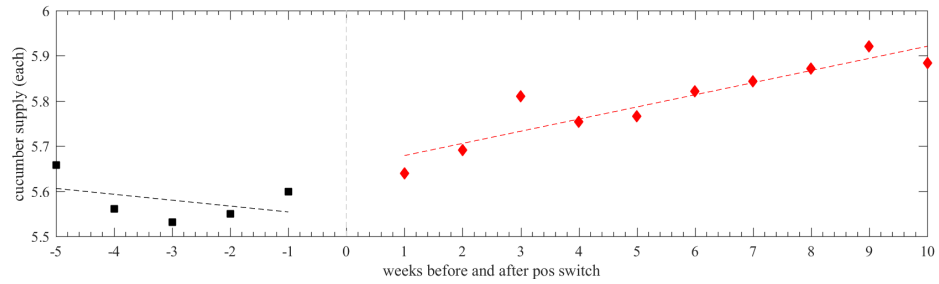
In Section 4, we highlight a surprising decline in customer satisfaction following the POS switch. This trend is puzzling, especially considering that a POS system upgrade provides the firm with real-time sales data for each store. Ideally, this should facilitate more efficient operations and boost customer satisfaction, as suggested unanimously in the literature. In this section, we delve deeper to uncover the underlying reasons for this anomaly. Firstly, we present model-free evidence and a regression discontinuity analysis to show that store managers order significantly larger quantities of ingredients post-POS switch. Given that the POS switch does not markedly influence demand, we propose our hypothesis that revolves around a misalignment between the IT upgrade and ingredient procurement. We fortify this theory with a range of corroborative evidence.

5.1. Impact of POS Switch on Ingredient Supply.

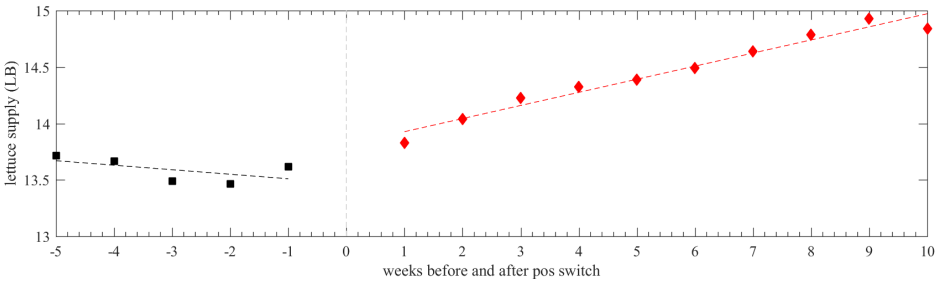
5.1.1. Model-Free Evidence. Figure 4 presents the average weekly order quantities for selected ingredients surrounding the times of the POS switch for all stores in the sample. In a manner similar to Figure 3, we realign the time periods based on the week each store transitioned to the new POS system. In this adjusted timeline, week 0 represents the week when the POS system was switched across all stores. A clear trend emerges from the data: The average weekly supply remains relatively stable (albeit with a slight decline for most ingredients) in the weeks leading up to the POS switch. However, there's a noticeable increase immediately after the transition, which then continues on an upward trajectory.

5.1.2. Regression Discontinuity Results. Table 6 summarizes the effects of the POS switch on ingredient supplies. These coefficients are estimated from the base model (i.e., Equation (1)), where the dependent variable is the average weekly supply of ingredients. Consistent with Figure 4, we observe a noticeable increase in ingredient supply post the POS switch, as evident from the positive coefficients linked to the POS implementation in the model. This increase is statistically significant across all ingredients.

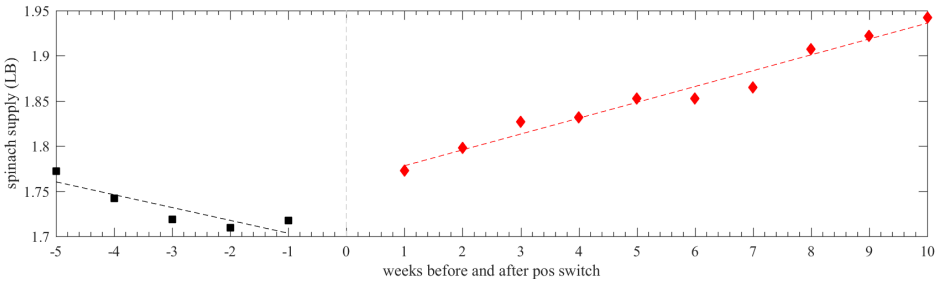
Figure 4 Ingredient order quantities during the weeks around POS switch.



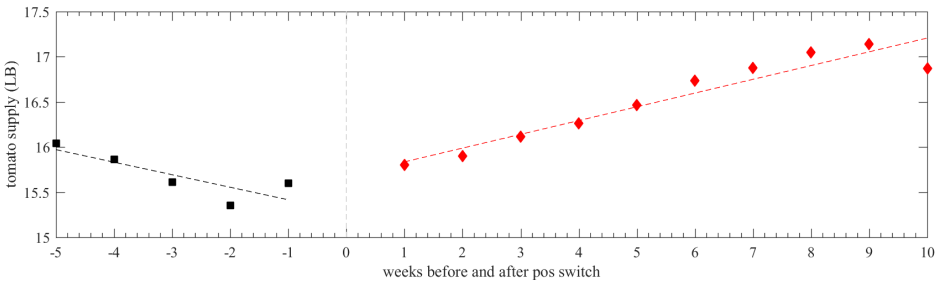
(a) cucumber



(b) lettuce



(c) spinach



(d) tomato

Note: The x-axis indicates the week. The y-axis denotes the average weekly order quantity of the corresponding ingredient across all stores in the sample. Dotted lines are fitted values from first-order polynomial regressions on either side of the time interval of the POS switch.

Table 6 Impact of POS switch on ingredients supply

	cucumber	lettuce	spinach	tomato
POS	0.096**	0.240***	0.062***	0.298**
	(0.048)	(0.091)	(0.014)	(0.118)
promotion	−0.244	−0.821***	−0.001	−0.149*
	(0.161)	(0.318)	(0.041)	(0.309)
opentime	−0.003**	−0.005***	−0.001***	−0.007***
	(0.001)	(0.002)	(0.000)	(0.003)
Month fixed effects	Yes	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes	Yes
Number of observations	35,810	35,810	35,810	35,810
R^2	0.754	0.749	0.868	0.773
Adj. R^2	0.726	0.721	0.854	0.747

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The regression analyses utilize data spanning 5 weeks prior to and 5 weeks following the POS switch.

Interestingly, it appears that store managers tend to order less during promotional periods. While this trend is not always statistically significant, it is noteworthy for ingredients like lettuce and tomato. This phenomenon could be attributed to a substitutional effect when new menu items are introduced during promotions. For instance, the company might promote a new guacamole-infused menu item during summer, which is not typically available throughout the year. Consequently, other ingredients, such as lettuce, will be substituted, leading to lower order quantities. The negative coefficient associated with *opentime* signifies a decline in per-store sales throughout the observed period.

5.2. Hypothesis and Supporting Evidence

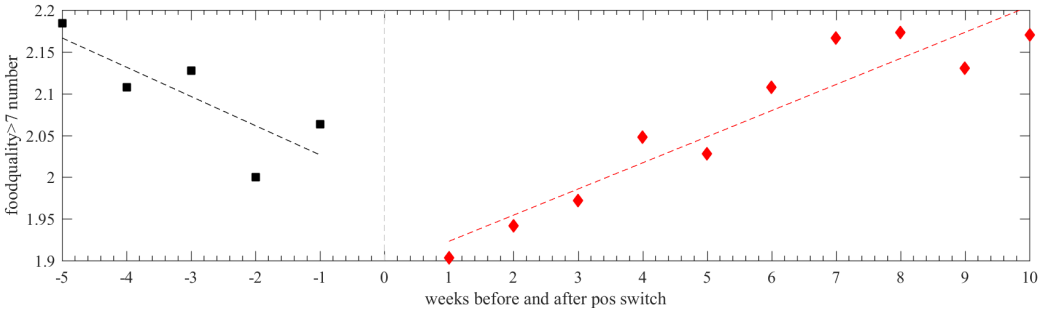
5.2.1. Hypothesis. So far, our analysis yields two primary observations: (1) There is a marked increase in the quantity of ingredients ordered by store managers immediately following the POS switch. (2) Customer satisfaction significantly declines post the POS switch. These observations collectively point towards a potential disconnect between the IT upgrade (specifically, the transition to a new POS system) and ingredient procurement. The company seemingly did not foresee the heightened demand from stores post the POS switch (the reasons for which will be detailed in Section 6) and consequently failed to adjust ingredient sourcing accordingly.

In our discussions with the firm's senior management, it was highlighted that "national chains rely on a network of suppliers set up to ship tomatoes, hamburger buns, and other staples in regularly scheduled truckloads. They are not always equipped for a sudden surge in demand for cookie-butter milkshakes or mango shrimp rice bowls." This unexpected spike in orders after the POS switch compelled the firm to expedite orders from suppliers, which likely compromised ingredient quality, leading to diminished customer satisfaction. In the following sections, we present a

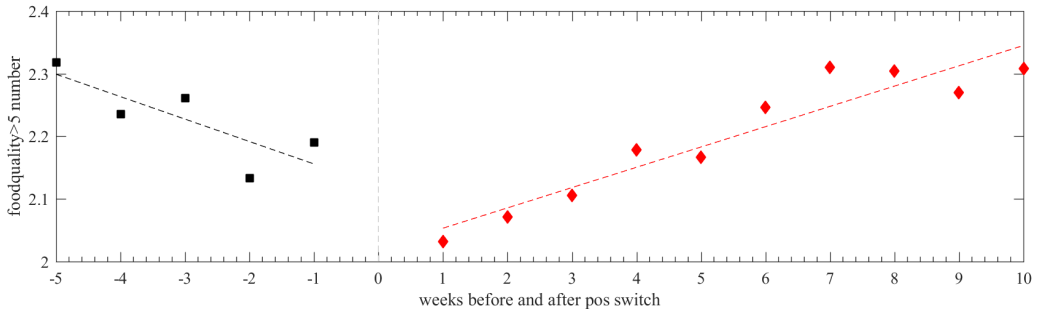
series of analyses to substantiate our hypothesis.

5.2.2. Customer Dissatisfaction on Food Quality Post-POS Switch. Numerous factors can contribute to customer dissatisfaction, including subpar food quality, extended wait times, unsanitary conditions, and more. Our hypothesis posits that the decline in ingredient quality, resulting in a less satisfactory end product, might be the driving force behind the decreased customer satisfaction post-POS switch. To validate this assumption, we delve into shifts in customer satisfaction specifically related to food quality around the times of the POS switch.

Figure 5 Customer satisfaction on food quality during the weeks around POS switch.



(a) Satisfaction on food quality > 7



(b) Satisfaction on food quality > 5

Note: The x-axis indicates the week. The y-axis in panel (a) (resp., panel (b)) denotes the average number of surveys with a satisfaction score on food quality > 7 (resp., > 5) per week. Dotted lines are fitted values from first-order polynomial regressions on either side of the time interval of the POS switch.

We first present model-free evidence to understand the impact of the POS switch on customer satisfaction on food quality. Similar to Figures 3 and 4, we realign the time periods in data by the week when the POS was switched for each store and plot the average number of surveys with a satisfaction score on food quality greater than 7 (resp., greater than 5) per week in Figure 5(a) (resp., Figure 5(b)). The pattern of customer satisfaction regarding food quality mirrors that of

overall satisfaction around the time of the POS switch. There is a noticeable decline leading up to, and immediately following, the POS transition, followed by an upward shift thereafter. Notably, dissatisfaction peaks during the first-week post-POS switch.

Table 7 Regression results on food satisfaction

	Food satisfaction > 7	Food satisfaction > 5
POS	−0.136***	−0.140***
	(0.046)	(0.047)
promotion	−0.047	−0.065
	(0.169)	(0.167)
opentime	−0.000	−0.000
	(0.001)	(0.001)
Month fixed effects	Yes	Yes
Store fixed effects	Yes	Yes
Number of observations	35,810	35,810
Number of stores	3,581	3,581
R^2	0.639	0.642
Adj. R^2	0.599	0.602

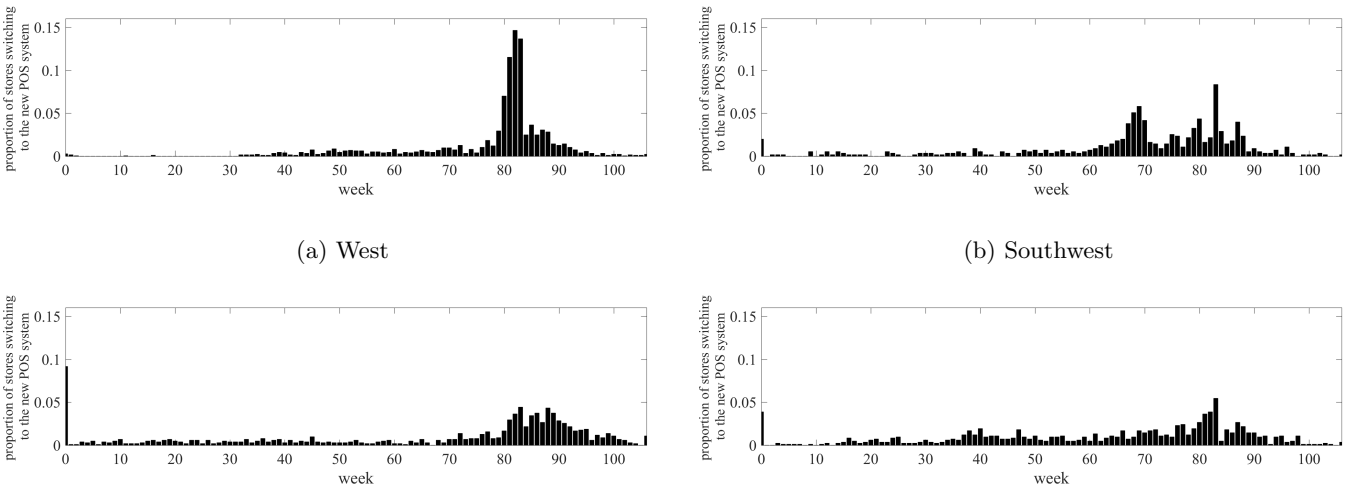
Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The regression analyses utilize data spanning 5 weeks prior to and 5 weeks following the POS switch.

The results from the regression, as detailed in Table 7, further underscore this observation. Table 7 showcases coefficients derived from Equation (1), where the dependent variable is the weekly average number of surveys yielding a food quality satisfaction score greater than 7 or 5. As evidenced by the table, post-POS switch, there is a decrease of 0.136 in the average number of surveys with satisfaction scores on food quality above 7 and a decrease of 0.140 for scores above 5. This discrepancy is statistically significant at the 1% level. Both the model-free evidence and regression outcomes indicate a deterioration in food quality following the POS switch.

5.2.3. Geographical Heterogeneity. Significant variability is evident in the timing of POS switches across different regions, as depicted in Figure 6. The x-axis represents the week, while the y-axis displays the percentage of stores in each geographical region transitioning to the new POS system within a specific week. A clear observation from the figure is the concentrated timing of POS switches in the west region, where nearly half of the stores transition between weeks 80 and 84. In contrast, the switch timings in other regions are more dispersed. Among these, the Midwest exhibits the most uniform spread of switches throughout the observation period.

Our hypothesis predicts that a concentrated surge in upgrades within a short span would likely exert added pressure on the supply chain, leading to reduced ingredient quality and a decline

Figure 6 Proportion of stores switching to the new POS system.



(a) West (b) Southwest

(c) Northeast (d) Midwest

Note: The x-axis indicates the week. The y-axis indicates the proportion of stores that switched to the new POS system from each geographical region within a certain week.

in customer satisfaction. Next, we verify this conjecture through a formal regression discontinuity analysis. To delve deeper into the varied impact of the POS switch on customer satisfaction across different contexts, we augment the base model to include interaction terms between store characteristics and the POS dummy. Specifically, the extended model adheres to the following specifications:

$$Y_{it} = \alpha_i + \beta POS_{it} + \boldsymbol{\theta} \boldsymbol{X}_{it} + \boldsymbol{\gamma} POS_{it} \cdot \boldsymbol{Z}_{it} + \epsilon_{it}. \tag{2}$$

Similar to the base model, α_i measures the store fixed effects. POS_{it} is a dummy variable which equals 1 if store i is using the new POS system in week t, and 0 otherwise. $\boldsymbol{X}_{it}$ includes the dummy variable $promotion_{it}$ and monthly dummies $month_t$.

The distinctive feature of the extended model is the inclusion of interaction terms that combine the POS dummy with various store characteristics, denoted as $\boldsymbol{Z}_{it}$. In particular, $\boldsymbol{Z}_{it}$ includes three regional dummies, i.e., $Southwest_i$, $Northeast_i$ and $Midwest_i$, which indicate the region within which store i is located. If all three regional dummies are set to zero, it indicates that the store is located in Western America. By incorporating these interaction terms, we aim to discern how the effects of the POS upgrade vary across different geographical regions. Beyond the geographical dummies, we also interact with the POS dummy variable with several other store characteristics, which provides deeper insights into the varied impacts of the POS upgrade. These variables include (1) $sale_i$, which indicates the average sales per week for store i. The calculation is based on the

average weekly sales following the transition to the new POS system, as sales data became available only after this change. (2) $type_i$ is a dummy variable indicating the store type as discussed in Table 1. $type_i$ equals 1 if store i is located within a larger business facility and 0 if it operates as a stand-alone retailer. (3) $loyalty_i$ is a dummy variable which equals 1 if store i has participated in the restaurant chain's loyalty program, and 0 otherwise. (4) $opentime_{it}$ indicates the number of days since its inception for store i at the beginning of week t. (5) $distance_i$ represents the shortest route length between store i and its primary distribution center, as determined by Google Maps.

The results of our extended model, focusing on the average number of surveys with an overall satisfaction score greater than 7, can be found in Column I of Table 8. Meanwhile, findings regarding the satisfaction score related to food quality are detailed in Column IV. Our estimates suggest a noticeable heterogeneity in the POS switch's impact across distinct geographic regions, as indicated by the interactions between the POS dummy and regional dummies. For context, the base group for both regressions is the Western United States region. When compared to this base group, we note a reduced decrease in the average number of surveys (with a satisfaction score above 7) per store weekly after the POS switch in both the Northeast and Midwest regions, with reductions of 0.180 and 0.359, respectively. This implies that stores in the Northeast and Midwest experience a more muted decline in customer satisfaction following the POS transition. Our observation that the effects of the POS switch vary across regions strongly corroborates our initial hypothesis. Specifically, in areas where the new POS system's implementation was more gradual, such as the Northeast and Midwest, the gap in customer satisfaction (defined by the difference in the average number of surveys with a score above 7 before and after the POS update) is significantly smaller. Further analysis, available in the Appendix, confirms that these insights remain consistent regardless of the chosen cutoff values.

Another noteworthy interaction term is between the POS dummy variable and the distance from store i to its primary distribution center. This is likely attributable to varied regional effects as well. Distance serves as a surrogate indicator for geographical factors due to differing population densities across regions.

5.2.4. Network Externality. Building upon the findings presented in Section 5.2.3, our hypothesis regarding the misalignment between IT enhancements and ingredient supply suggests another dimension. Specifically, there could be an inverse correlation between the decline in customer satisfaction concerning food quality after the POS switch and the number of stores undergoing the POS transition simultaneously within a week. A surge in stores adopting the new POS system during the same timeframe can intensify pressures on the supply chain. This, in turn,

Table 8 Regression results on overall customer satisfaction and satisfaction on food quality

	Overall satisfaction > 7			Food quality > 7		
	I	II	III	IV	V	VI
POS	−0.211*	0.060	−0.018	−0.250*	0.073	−0.014
	(0.124)	(0.050)	(0.128)	(0.124)	(0.050)	(0.128)
promotion	0.004	−0.020	−0.004	−0.043	−0.070	−0.051
	(0.160)	(0.161)	(0.158)	(0.165)	(0.166)	(0.163)
opentime	−0.001	−0.002**	−0.002**	−0.001	−0.003***	−0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
POS × sale	−1.3e-5		−1.0e-5	−1.1e-5		−9.0e-6
	(8.8e-6)		(8.8e-6)	(8.7e-6)		(8.7e-6)
POS × Southwest	0.076		−0.036	0.074		−0.037
	(0.103)		(0.108)	(0.104)		(0.109)
POS × Northeast	0.180**		0.098	0.173**		0.092
	(0.071)		(0.076)	(0.071)		(0.076)
POS × Midwest	0.359***		0.210***	0.357***		0.211***
	(0.066)		(0.076)	(0.067)		(0.076)
POS × type	−0.070		−0.078	−0.063		−0.070
	(0.084)		(0.084)	(0.085)		(0.085)
POS × loyalty	−0.099		−0.057	−0.108		−0.068
	(0.067)		(0.067)	(0.067)		(0.067)
POS × opentime	4.0e-7		8.6e-7	1.3e-6		1.8e-6
	(8.6e-6)		(8.6e-6)	(8.7e-6)		(8.7e-6)
POS × distance	0.002***		0.001**	0.002***		0.001**
	(0.000)		(0.000)	(0.000)		(0.000)
posnum		−3.7e-4	−7.3e-4*		−3.1e-4	−6.6e-4*
		(3.5e-4)	(4.0e-4)		(3.5e-4)	(4.1e-4)
POS × posnum		−0.003***	−0.002***		−0.003***	−0.002***
		(0.000)	(0.000)		(0.000)	(0.000)
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	35,810	35,810	35,810	35,810	35,810	35,810
Number of stores	3,581	3,581	3,581	3,581	3,581	3,581
R^2	0.639	0.639	0.640	0.640	0.640	0.641
Adj. R^2	0.598	0.599	0.599	0.600	0.600	0.600

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; $posnum_{it}$ indicates the number of stores switching POS within the same geographical region as store i in week t.

may result in compromised ingredient quality, culminating in decreased customer satisfaction. To empirically test this prediction, we introduce a new variable, $posnum_{it}$. This variable quantifies the number of stores that transitioned to the new POS system within the same geographical region as store i during week t. Our analysis of this potential network externality is confined to the regional level, given that ingredient suppliers are usually unique to distribution centers within specific regions rather than being shared across multiple regions.

To conduct our analysis, we include $posnum_{it}$ and its interactive term $POS \times posnum_{it}$ as

added independent variables in the extended model. We execute this both with and without the inclusion of other interaction terms. The outcomes of this analysis are consolidated and presented in Table 8. In the regressions where the dependent variable represents the number of surveys with an overall satisfaction score exceeding 7, i.e., Columns II and III of Table 8, the coefficient for the standalone variable *posnum* is negative and statistically insignificant in the model without other interaction terms. However, when all control variables are considered, its significance improves, albeit marginally. On the other hand, the coefficients for the interaction term $POS \times posnum$ are notably larger in magnitude and robustly significant at the 1% level across both models. Similar patterns are evident when we pivot our focus to the regressions that consider the number of surveys with food quality satisfaction scores above 7, as displayed in Columns V and VI of Table 8.

Several insights warrant discussion: (1) The significance of the interaction term $POS \times posnum$ underscores that the volume of stores transitioning to a new POS system in a specific week can indeed influence the customer satisfaction of a focal store, especially if that store has recently implemented the POS upgrade itself. This negative network externality speaks directly to the supply challenges at the regional level stemming from the POS transition. (2) The supply challenge appears more pronounced for stores that have just updated their POS systems. Possibly, this discrepancy might be attributed to the company's inventory management strategy. Routine orders might receive precedence and be serviced through standard channels. In contrast, surges in demand post-POS upgrades might be addressed using alternative suppliers. These alternative channels might not subject their ingredients to the rigorous evaluations expected from certified suppliers.

5.2.5. Longer Term Effects. If our hypothesis regarding the misalignment between IT enhancements and ingredient supply holds true, the adverse effect of the POS upgrade should be transitory. This is because, in response to observing increased order quantities from the stores, the firm would likely seek alternative supply sources. As a result, the supply situation would improve over time, leading to a rise in customer satisfaction.

To test this prediction, we explore the long-term evolution of customer satisfaction following the POS upgrade. Specifically, we utilize 15 weeks of data surrounding the POS transition for our regression, encompassing 5 weeks prior to the switch and 10 weeks post-switch. We substitute the POS dummy variable with two distinct binary variables: $Weeks_{1-5}$ and $Weeks_{6-10}$, representing the initial five weeks and the subsequent five weeks after the POS transition, respectively. The coefficients of these binary variables gauge the disparity in customer satisfaction relative to the 5-week period preceding the POS switch, which is designated as the reference group.

Table 9 Impact of POS switch on customer satisfaction in the longer term

	Overall satisfaction > 7	Food quality > 7
Weeks 1–5 (post POS switch)	-0.148^{***} (0.041)	-0.143^{***} (0.041)
Weeks 6–10 (post POS switch)	-0.041 (0.064)	-0.027 (0.064)
Promotion	-0.095 (0.107)	-0.125 (0.113)
Open time	-0.001 (0.001)	-0.001 (0.001)
Month fixed effects	Yes	Yes
Store fixed effects	Yes	Yes
Number of observations	53,412	53,412
R^2	0.611	0.610
Adj. R^2	0.583	0.582
	Test for $H_0 : \beta_{Weeks\ 1-5} = \beta_{Weeks\ 6-10}$	
p value	0.002	0.001

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. 15 weeks of data surrounding the POS transition are used for the regressions, encompassing 5 weeks prior to the switch and 10 weeks post-switch. $Weeks_{1-5}$ and $Weeks_{6-10}$ represent the initial five weeks and the subsequent five weeks after the POS transition, respectively.

The findings are detailed in Table 9. Notably, the coefficients for the $Weeks_{1-5}$ dummy are negative and exhibit strong statistical significance at the 1% level in both models. This aligns with our earlier observations from Tables 5 and 7, reinforcing the notion of a decline in customer satisfaction following the POS switch. However, there is an evident uptick in customer satisfaction during the subsequent period, spanning from the 6_{th} to the 10_{th} week after the POS transition. The non-significance of the coefficients for $Weeks_{6-10}$ indicates that customer satisfaction levels during this timeframe are comparable to those observed in the 5-week window preceding the POS upgrade. Furthermore, the bottom half of Table 9 reveals a marked improvement in customer satisfaction during the second 5-week interval post-POS transition when contrasted against the immediate 5-week period following the upgrade. These results further reinforce our hypothesis regarding the misalignment between IT enhancements and ingredient supply. Following the initial surge in the store order quantities after the POS upgrade, the firm adjusted by steadily bolstering the supply of agricultural products. Over time, this adaptability would alleviate the initial downturn in customer satisfaction post-POS upgrade.

6. Why Order More After POS Switch?

As demonstrated by the model-free evidence in Figure 4 and the regression discontinuity findings in Table 6, store managers significantly increase ingredient orders following a POS upgrade. This leads

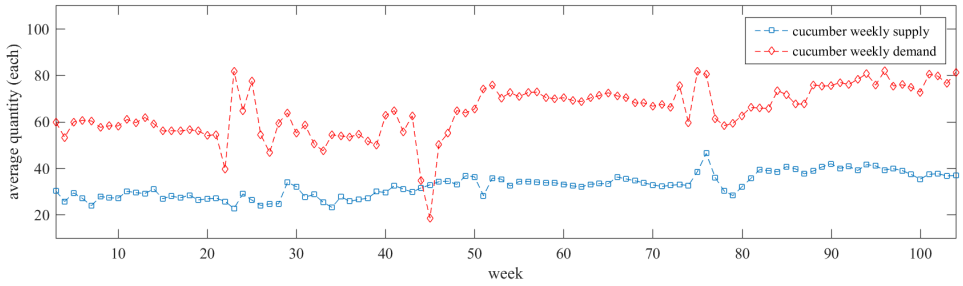
to the following question: Why do order quantities see such a marked rise post-POS transition? We put forth two hypotheses to elucidate this phenomenon. One hypothesis is rooted in algorithmic changes, while the other is grounded in human behavior. We assess these hypotheses against our dataset to identify which aligns most closely with the observed patterns.

First, from an algorithmic standpoint, the surge in order quantities might be attributed to a reduced information lead time following the POS upgrade. Most store managers place orders for ingredient supplies by following recommendations from the company headquarter. These recommendations are generated by a back-end algorithm. At its core, the algorithm solves a data-driven newsvendor problem. With the enhanced POS system, this algorithm now has access to real-time demand data, as opposed to the 10-day-old data from the previous system. This immediacy substantially diminishes uncertainty in demand forecasting. With a reduction in the variance of predicted demand, it is well known in the literature that the direction of change in the newsvendor order quantity hinges on the critical fractile (Chopra 2019). Specifically, if the critical fractile is less than 0.5, the optimal order quantity will increase, eventually converging to the mean as the demand uncertainty decreases. Conversely, if the critical fractile exceeds 0.5, the optimal order quantity will decrease with diminishing demand uncertainty.

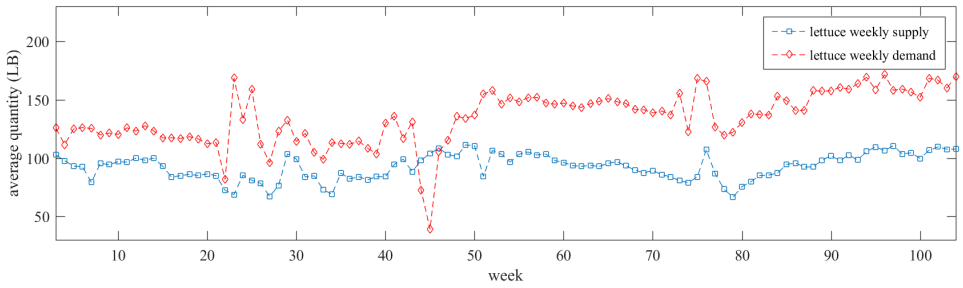
Generally, pinpointing the exact value of the critical fractile can be challenging because sales data is censored. However, in our specific context, the company provides guidance on the quantity of ingredients required to craft each customized food product. As a result, by coupling this guidance with the sales data, we can deduce the theoretical ingredient demand within each cycle, and the ratio of actual supply to theoretical demand would shed some light on the value of the critical fractile. Figure 7 displays the results for four selected ingredients. In the figure, the blue line represents the average weekly supply, while the red line indicates the average weekly (theoretical) demand for each corresponding ingredient. It is evident from the graph that the supply level persistently trails the theoretical demand across all four ingredients, suggesting a critical fractile value below 0.5. This observation aligns with established literature recommendations on inventory levels for perishable products in a newsvendor setting (Krishnan and Winter 2010).

Building on the prior discussion, the accessibility of real-time demand information should lead to an uptick in optimal order quantities, resulting in elevated order levels post-POS upgrade. Moreover, if this pronounced increase in order quantities is indeed driven by the shortened information lead time, we would anticipate that this heightened supply level is sustained subsequent to the upgrade. We summarize this hypothesis as follows.

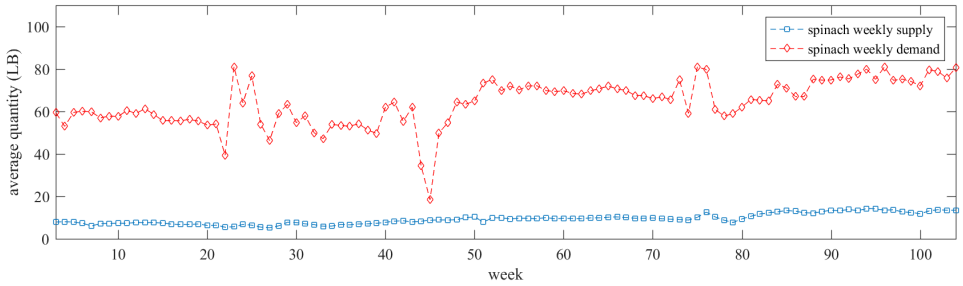
Figure 7 Ingredient supply vs. theoretical demand.



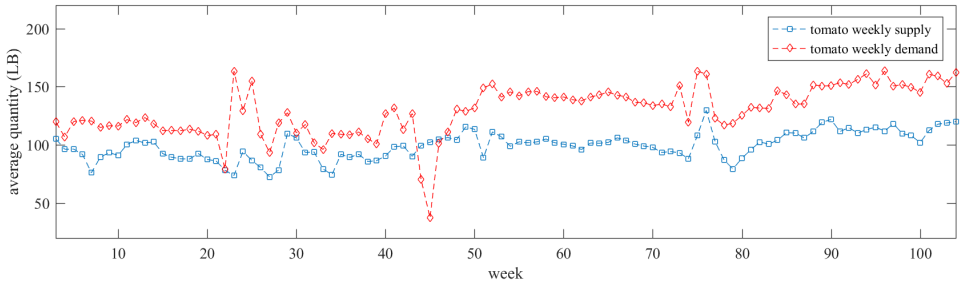
(a) cucumber



(b) lettuce



(c) spinach



(d) tomato

Note: The x-axis indicates the week. The blue line represents the average weekly supply, while the red line indicates the average weekly (theoretical) demand for each corresponding ingredient.

HYPOTHESIS 1. *The elevated order quantities are attributed to the reduction of information lead time. In this case, the post-POS-upgrade levels would be sustained throughout.*

Alternatively, the increased order quantities post-POS upgrade could be a manifestation of strategic behavior by store managers. These managers are naturally inclined to ensure a seamless and scheduled ingredient supply. Yet, in light of potential disruptions anticipated during the POS upgrade, they might deliberately increase their orders as a safeguard against this uncertainty. If this surge in order quantities is primarily a result of such strategic behavior, one would expect order levels to revert to their pre-POS-upgrade baseline over time. This observation gives rise to our second hypothesis.

HYPOTHESIS 2. *The elevated order quantities are attributed to store managers' strategic behavior. In this case, the post-POS-upgrade levels would revert to their pre-POS-upgrade baseline over time.*

Understanding the underlying reasons for the increased order quantities from stores post-POS upgrade is pivotal for the firm, as differing reasons call for distinct interventions. Should the real-time demand information be the primary catalyst for the elevated order volumes, the most appropriate response would be to preemptively stockpile ingredients. Conversely, if the increased order quantities stem from store managers' strategic maneuvering, the stress imposed on the supply chain is entirely preventable. This can be mitigated, at least partially, through enhanced communication between the headquarter and the stores, emphasizing procedural clarity, contingency plans, and related matters.

Next, we seek to understand the longer-term impact of the POS switch on ingredient supply, with the goal of determining which hypothesis garners support from the data. Similar to our methodology in Section 5.2.5, we employ data spanning 5 weeks prior to and 10 weeks post-POS switch to gauge its enduring consequences on ingredient supply. The specification is the same as the base model, except that the POS dummy variable is replaced by two distinct binary variables: $Weeks_{1-5}$ and $Weeks_{6-10}$, which represent the initial five weeks and the subsequent five weeks after the POS transition, respectively. The dependent variable is the average weekly supply of the chosen quartet of ingredients: cucumber, lettuce, spinach, and tomato.

Table 10 illustrates a persistent increase in the average weekly supply following the POS transition. The order quantities for all four ingredients are notably higher during both the initial five-week period and the subsequent five-week period after the POS upgrade. This trend is evident from the positive and statistically significant coefficients associated with the $Weeks_{1-5}$ and $Weeks_{6-10}$

Table 10 Impact of POS switch on ingredient supply in the longer term

	cucumber	lettuce	spinach	tomato
Weeks 1-5	0.090**	0.184**	0.026**	0.281***
(post POS switch)	(0.041)	(0.080)	(0.011)	(0.101)
Weeks 6-10	0.114*	0.276**	0.044**	0.674***
(post POS switch)	(0.068)	(0.130)	(0.019)	(0.165)
Promotion	−0.465***	−1.083***	−0.070***	−0.414*
	(0.122)	(0.283)	(0.021)	(0.252)
Open time	−0.001*	−0.003*	−0.000	−0.006***
	(0.000)	(0.002)	(0.000)	(0.002)
Month fixed effects	Yes	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes	Yes
Number of observations	53,415	53,423	53,333	53,420
R^2	0.747	0.742	0.871	0.767
Adj. R^2	0.729	0.724	0.861	0.750
		Test for $H_0 : \beta_{Weeks1-5} = \beta_{Weeks6-10}$		
p value	0.485	0.173	0.064	0.000

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. 15 weeks of data surrounding the POS transition are used for the regressions, encompassing 5 weeks prior to the switch and 10 weeks post-switch. $Weeks_{1-5}$ and $Weeks_{6-10}$ represent the initial five weeks and the subsequent five weeks after the POS transition, respectively.

dummy variables across all regression models. Intriguingly, ingredient supply levels tend to be even higher during the latter five-week span, as suggested by the test results at the bottom of Table 10. Such a trend might be attributable to the progressive enhancements on the supply front, with the firm steadily onboarding alternative suppliers to accommodate the amplified demand from stores. Collectively, the evidence presented in Table 10 lends credence to Hypothesis 1, positing that the surge in order quantities post-POS transition is primarily driven by the reduction of information lead time.

Similar to Table 6, the significance of the promotion variable might be attributable to a substitution effect observed when new menu items are launched during promotions. The negative coefficient linked with *opentime* indicates a decreasing trend in per-store sales over the observation period.

Interestingly, this result also speaks to the ongoing discussion about the interplay between algorithm and human decision-making. With the new POS system reducing information lead times and prompting higher order quantities, store managers inclined to rely on algorithmic recommendations, moving away from pre-POS-upgrade ordering quantities. Although these algorithm-based recommendations appear optimal under the new information regime, a lack of necessary adjustments on the supply side by the firm leads to diminished customer satisfaction, ultimately resulting in less-than-ideal outcomes.

7. Conclusion

In our study, we delve into the effects of a POS system upgrade for a major quick-service restaurant chain in the U.S. While the modernized system, with a shortened information lead time, enhances the algorithm's capability to forecast demand with increased precision—a commendable advancement in itself—it also gives rise to unforeseen challenges. With the more accurate demand prediction comes a surge in the recommended order quantities, owing to the reduced uncertainty surrounding demand. Regrettably, the company seemed unprepared for the heightened demand from stores following the POS upgrade, leading to an oversight in adjusting its ingredient sourcing strategy in response to these changes.

This challenge transcends the specific firm in question. In the current era, the profound impact of information technology advancements on supply chain management, especially in terms of information sharing, is undeniable. Numerous companies are actively exploring and integrating cutting-edge techniques like blockchain technology and the Internet of Things into their daily processes. Our findings underscore the complexities inherent in the digital transformation. For organizations to fully harness the benefits of these technological innovations, a concerted and synchronized approach is essential, involving close collaboration between different departments, including IT and operations. This integrated effort is crucial for capitalizing on the transformative power of these digital tools in the supply chain landscape.

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Online Appendix: Robustness Checks

Impact of POS upgade on customer satisfaction

Similar to Tables 5 and 7, results in Table A.1 are estimated based on the base model, as shown in Equation (1), using the data within 5 weeks before and 5 weeks after POS upgrade. In the analysis, the dependent variables are the average customer satisfaction scores and the average scores related to food quality. Echoing our primary findings, both metrics exhibit a decline post the POS system upgrade. The decrease in the average customer satisfaction scores is statistically significant at the 5% level. However, the dip in scores centered around food quality is statistically insignificant. As articulated in the main body of the text, quantitatively interpreting customer satisfaction ratings presents challenges. Consequently, in our main analysis, we favor the frequency of these scores over their average to gauge customer satisfaction.

Table A.1 Impact of POS switch on average customer satisfaction scores

	Overall satisfaction	Satisfaction on food quality
POS	-0.114^{**}	-0.027
	(0.050)	(0.039)
Promotion	0.334	0.045
	(0.263)	(0.268)
Open time	0.001	-0.000
	(0.001)	(0.000)
Month fixed effects	Yes	Yes
Store fixed effects	Yes	Yes
Number of observations	25,692	25,692
R^2	0.247	0.252
Adj. R^2	0.125	0.131

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The regression analyses utilize data spanning 5 weeks prior to and 5 weeks following the POS switch.

Geographical heterogeneity and network externality

Table A.2 is the same as Table 8 in Section 5.2, except that we now adopt a satisfaction score threshold of 5. As shown in the table, the results are robust with respect to the choice of cutoff, and all major conclusions still hold qualitatively with a cutoff of 5.

Longer term effects

Table A.3 is the same as Table 9 in Section 5.2.5, except that we now adopt a satisfaction score threshold of 5. In alignment with our observations from Table 9, there is a notable decline in customer satisfaction immediately following the POS upgrade. This trend is evidenced by the significant negative coefficient associated with the $Weeks_{1-5}$ dummy variable. However, over time,

Table A.2 Regression results on overall satisfaction and satisfaction on food quality (with a cutoff of 5)

	Overall satisfaction > 5			Food quality > 5		
	I	II	III	IV	V	VI
POS	−0.215*	0.056	−0.025	−0.233*	0.074	−0.035
	(0.128)	(0.052)	(0.132)	(0.129)	(0.052)	(0.133)
promotion	−0.025	−0.052	−0.035	−0.059	−0.088	−0.069
	(0.161)	(0.163)	(0.160)	(0.163)	(0.164)	(0.162)
opentime	−0.001	−0.002**	−0.002**	−0.001	−0.003***	−0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
POS × sale	−1.4E-5		−1.1E-5	−1.1E-5		−9.1E-6
	(9.2E-6)		(9.1E-6)	(9.1E-6)		(9.1E-6)
POS × Southwest	0.058		−0.052	0.050		−0.065
	(0.106)		(0.111)	(0.106)		(0.111)
POS × Northeast	0.181**		0.102	0.179**		0.096
	(0.073)		(0.078)	(0.074)		(0.079)
POS × Midwest	0.365***		0.218***	0.362***		0.210***
	(0.068)		(0.077)	(0.069)		(0.078)
POS × type	−0.081		−0.089	−0.074		−0.082
	(0.086)		(0.086)	(0.087)		(0.087)
POS × loyalty	−0.104		−0.063	−0.105		−0.062
	(0.069)		(0.069)	(0.070)		(0.070)
POS × opentime	2.1E-6		2.5E-6	3.3E-6		3.7E-6
	(8.8E-6)		(8.8E-6)	(8.9E-6)		(8.9E-6)
POS × distance	0.002***		0.001***	0.002***		0.001***
	(0.000)		(0.000)	(0.000)		(0.000)
posnum		−4.2E-4	−7.9E-4*		−3.9E-4	−7.3E-4*
		(3.6E-4)	(4.2E-4)		(3.7E-4)	(4.2E-4)
POS × posnum		−0.003***	−0.002***		−0.003***	−0.002***
		(0.000)	(0.000)		(0.000)	(0.000)
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	35,810	35,810	35,810	35,810	35,810	35,810
Number of stores	3,581	3,581	3,581	3,581	3,581	3,581
R^2	0.644	0.644	0.645	0.643	0.643	0.644
Adj. R^2	0.604	0.605	0.605	0.603	0.604	0.604

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; $posnum_{it}$ indicates the number of stores switching POS within the same geographical region as store i in week t.

customer satisfaction appears to recover, indicated by the statistically insignificant coefficient of the $Weeks_{6-10}$ dummy variable.

Table A.3 Impact of POS switch on customer satisfaction in the longer term (with a cutoff of 5)

	Overall satisfaction > 5	Food quality > 5
Weeks 1–5 (post POS switch)	-0.152^{***}	-0.144^{***}
	(0.042)	(0.042)
Weeks 6–10 (post POS switch)	-0.049	-0.028
	(0.066)	(0.066)
Promotion	-0.101	-0.157
	(0.116)	(0.119)
Open time	-0.001	-0.001
	(0.000)	(0.000)
Month fixed effects	Yes	Yes
Store fixed effects	Yes	Yes
Number of observations	53,412	53,412
R^2	0.614	0.614
Adj. R^2	0.587	0.586
	Test for $H_0 : \beta_{Weeks\ 1-5} = \beta_{Weeks\ 6-10}$	
p value	0.003	0.001

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. 15 weeks of data surrounding the POS transition are used for the regressions, encompassing 5 weeks prior to the switch and 10 weeks post-switch. $Weeks_{1-5}$ and $Weeks_{6-10}$ represent the initial five weeks and the subsequent five weeks after the POS transition, respectively.